# Γ-Net: Minimum Reflection Physics for Dual-Network Life

**Framework Definition, Quantity Inventory, and Verification Matrix**

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This paper defines the complete axiomatic framework of the Γ-Net (*Gamma-Net*) theory: a physics-driven model of biological organisation based on one variational principle—Minimum Reflection Action—and three inviolable constraints C1–C3. We provide: (i) the formal axioms and all derived quantities ($\Gamma$, $A_{\rm imp}$, $A_{\rm cut}$, $D_Z$, $H$, lifecycle ODE, $\Phi_{\rm meta}$); (ii) a unified verification matrix classifying every prediction as *internal consistency*, *literature-supported*, *NHANES-validated*, or *untested*; (iii) a falsifiable prediction roadmap for future experimental work. The framework unifies phenomena across 15 orders of magnitude in spatial scale and 29 tissue types under a single equation: $\Delta Z = -\eta\,\Gamma\,x_{\rm in}\,x_{\rm out}$. This paper contains no new results; it serves as the master reference for the five companion papers that develop and test the theory.

## I. INTRODUCTION: WHY THIS PAPER EXISTS

The Γ-Net framework has been developed across five companion papers, each deriving a distinct class of biological phenomena from the Minimum Reflection Principle (MRP). Paper 1 derives topology and metabolic scaling [1]; Paper 2 derives dual neural–vascular organ health [2]; Paper 3 derives sleep, the lifecycle equation, and aging [3]; Paper 4 derives memory, consciousness, and nociception [4]; Paper 5 validates the framework against NHANES epidemiological data and 29 tissue remodeling laws [5].

A reader entering at any single paper encounters a local recap of the axioms but lacks:

1. a single, canonical list of *all* defined quantities and their inter-relations;
2. a structured table distinguishing which claims have external (non-circular) support and which remain internal consistency checks;
3. a roadmap of falsifiable predictions organised by experimental modality.

This paper fills those three gaps. It contains no new derivations or simulations; every result cited here is proven elsewhere and cross-referenced to the relevant companion paper.

## II. THE THREE AXIOMS

The theory rests on one variational principle and three local constraints, labelled **C1–C3** (**C = Constraint**; the numbering parallels the Three Laws of Thermodynamics by design).

### A. The Minimum Reflection Principle (MRP)

**Axiom 1** (Minimum Reflection Action)**.** *Any impedance network subject to sustained energy flow evolves so as to minimise the total reflected action:*

$$\mathcal{A}[\Gamma] = \int_0^T \sum_i \Gamma_i^2(t)\,P_{in,i}(t)\,dt \;\to\; \min\,, \qquad (1)$$

*where $\Gamma_i = (Z_{load,i} - Z_{source,i})/(Z_{load,i} + Z_{source,i})$ is the reflection coefficient at interface $i$, and $P_{in,i}$ is the incident power.*

The MRP is not a modelling choice; it is the unique variational formulation consistent with the First Law of Thermodynamics at a one-dimensional interface (Paper 5, Section 2 [5]).

### B. C1 — Energy Conservation

**Axiom 2** (C1: Energy Conservation)**.** *At every interface $i$, at every instant $t$:*

$$\Gamma_i^2(t) + T_i(t) = 1\,, \qquad (2)$$

*where $T_i = 1-\Gamma_i^2$ is the transmission coefficient (fraction of incident power delivered to the load).*

C1 is the First Law of Thermodynamics applied locally. It is not an approximation; it holds exactly for any lossless interface. Dissipative corrections (Johnson–Nyquist noise, Arrhenius aging) modify $Z$ over time but never violate power balance at a single interface.

**Remark 1** (Signal power vs. metabolic work)**.** *C1 governs the instantaneous* signal power *budget at each interface: incident = reflected + transmitted. The metabolic energy that fuels C2 remodeling ($\Delta Z = -\eta\,\Gamma\,x_{in}\,x_{out}$)*

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*enters as a separate thermodynamic work term that modifies $Z$ itself, not the $\Gamma^2 + T = 1$ identity. Biological processes that actively consume ATP—receptor phosphorylation, immune affinity maturation, synaptic long-term potentiation—change the impedance landscape (and hence future values of $\Gamma$) but do not violate C1 at any instant. The analogy is an electronic amplifier: the DC power supply changes the gain (modifies $Z$), yet Kirchhoff's power law holds at every RF port.*

**Corollary 1** (Thermal Dissipation)**.** *At every interface where $\Gamma_i \neq 0$, the reflected fraction $\Gamma_i^2$ of incident power is not "lost"—it is re-absorbed by the source impedance and ultimately dissipated as Joule heat. For an organism with $N$ active interfaces, the total thermal dissipation rate is*

$$P_{\text{heat}} = \sum_{i=1}^{N} \Gamma_i^2 \, P_{\text{in},i} \,, \tag{3}$$

*where $P_{\text{in},i}$ is the incident power at interface $i$. This heat is not a by-product; it is the* thermodynamic cost of computation. *The body temperature $T_{\text{body}}$ in thermal steady state satisfies $P_{\text{heat}} = hA\,(T_{\text{body}} - T_{\text{env}})$, where $hA$ is the effective thermal conductance to the environment. The same impedance mismatches that perform neural computation therefore* generate the heat that sustains the temperature at which those computations remain viable *(Papers 3–4 [3, 4]).*

### C. C2 — Impedance Remodeling

**Axiom 3** (C2: Impedance Remodeling)**.** *The gradient descent of $\mathcal{A}$ with respect to local impedance yields:*

$$\Delta Z_i = -\eta \, \Gamma_i \, x_{in,i} \, x_{out,i} \,, \tag{4}$$

*where $\eta > 0$ is the remodeling rate, $x_{in,i}$ is the input signal (effort or flow arriving at interface $i$), and $x_{out,i}$ is the output signal (transmitted through interface $i$).*

Equation (4) is the unique local update rule that decreases $\mathcal{A}$ at every step (Paper 5, Section 2 [5]). Its form is identical to Hebb's learning rule when $Z$ is synaptic conductance, to Wolff's law when $Z$ is bone stiffness, to Glagov's remodeling when $Z$ is vascular compliance, and to 26 further tissue-specific remodeling laws (Table V).

**Historical note.** In the original formulation (Papers 1–4), C2 was called the "Hebbian update" and its signals were labelled $x_{\text{pre}}$, $x_{\text{post}}$. Paper 5 proved that the same equation governs all 29 tissue types examined, making the neural-specific terminology misleading. We adopt the generalised names *impedance remodeling* and $(x_{\text{in}}, x_{\text{out}})$ throughout this series.

### D. C3 — Impedance-Tagged Transport

**Axiom 4** (C3: Impedance-Tagged Transport)**.** *Every energy or material flow across an interface carries $(Z_{source}, Z_{load})$ as metadata, so that $\Gamma_i$ is locally computable without global network knowledge.*

C3 is a bookkeeping requirement, not an additional physical law. It states that whenever a signal crosses an impedance boundary, the downstream node can compute $\Gamma$ from local information alone. In biological terms: blood carries its viscosity ($Z_{\text{blood}}$) into a vessel of compliance ($Z_{\text{vessel}}$); a nerve impulse carries $R_m + 1/j\omega C_m$ into a synapse of input impedance $Z_{\text{syn}}$; a mechanical load carries force/velocity into a bone of stiffness $Z_{\text{bone}}$.

**Implementation note.** In the open-source reference implementation (ALICE), C3 is enforced by requiring all inter-module values to be `ElectricalSignal` objects carrying $Z$ metadata. Raw floating-point numbers between modules are forbidden at the code level.

## III. DERIVED QUANTITIES

Table I lists every named quantity in the Γ-Net framework, its defining equation, physical interpretation, and the companion paper where it first appears.

## IV. SERIES ARCHITECTURE

The five companion papers form a logical chain:

1. **Paper 1: Topology and Fractal Cost** [1]. Proves the irreducibility theorem ($\mathcal{A} = A_{\text{imp}} + A_{\text{cut}}$, $A_{\text{cut}} > 0$), derives relay-node hierarchies, the Murray bifurcation ratio $\beta = 2^{1/3}$, Kleiber's 3/4-power metabolic scaling, joint impedance transformers, and dual-network packing overhead $\Omega$.

2. **Paper 2: Dual-Network Organ Health** [2]. Applies MRP to vascular networks; derives Murray's law from Γ-minimisation; defines the dual-network health index $H = (1-\Gamma_n^2)(1-\Gamma_v^2)$; models five vascular diseases as impedance failure modes. Validated against 12/12 published clinical benchmarks with 100% pass rate.

3. **Paper 3: Temporal Dynamics** [3]. Introduces impedance debt $D_Z$; derives sleep as $D_Z$-discharge; derives the lifecycle equation governing cognition from birth to death; models emotion as Γ-dynamics readout; derives aging as Arrhenius fatigue; explains aging pain immunity via slow $Z$-drift.

4. **Paper 4: Memory, Consciousness, Nociception** [4]. Defines memory as $dZ/dt$, consciousness as closed-loop $\Phi_{\text{meta}}$, and the null space $\mathcal{S}$ ("soul") as $\ker(\partial Z/\partial t)$. Proves the Television Theorem distinguishing open-loop from closed-loop systems.

TABLE I. Complete quantity inventory of the Γ-Net framework.

| Symbol | Name | Definition | Interpretation | Paper |
|---|---|---|---|---|
| $\Gamma_i$ | Reflection coefficient | $\frac{Z_{\text{load},i}-Z_{\text{source},i}}{Z_{\text{load},i}+Z_{\text{source},i}}$ | Mismatch at interface $i$ | 0 |
| $T_i$ | Transmission coefficient | $1-\Gamma_i^2$ | Fraction of power delivered | 0 |
| $\mathcal{A}$ | Reflection action | $\int_0^T \sum_i \Gamma_i^2 P_{\text{in},i}\, dt$ | Total wasted energy | 0 |
| $A_{\text{imp}}$ | Improvable action | $\mathcal{A} - A_{\text{cut}}$ | Reducible by C2 remodeling | 1 |
| $A_{\text{cut}}$ | Irreducible (cutoff) action | $A_{\text{cut}} = \sum_{k>K^*} \Gamma_k^2$ | Topological cost immune to C2 | 1 |
| $D_Z$ | Impedance debt | $\int_0^{T_{\text{wake}}} \lvert\Gamma\rvert^2 P_{\text{in}}\, dt$ | Cumulative mismatch; cleared by sleep | 3 |
| $D_n$, $D_v$ | Neural / vascular debt | $D_Z = D_n + D_v$ | Dual-network decomposition | 3 |
| $H$ | Health index | $\prod_i (1-\Gamma_i^2)$ | Whole-body transmission efficiency | 5 |
| $\Sigma\Gamma^2$ | Total mismatch energy | $\sum_i \Gamma_i^2(t)$ | Instantaneous network mismatch | 3 |
| Lifecycle ODE | Lifecycle equation | $\frac{d(\Sigma\Gamma^2)}{dt} = -\eta\,\Sigma\Gamma^2 + \gamma\,\Gamma_{\text{env}} + \delta\, D(t)$ | Cognition bathtub curve | 3 |
| $\Phi_{\text{meta}}$ | Meta-reflection coefficient | $\frac{Z_{\text{obs}}-Z_{\text{act}}}{Z_{\text{obs}}+Z_{\text{act}}}$ | Self-monitoring; consciousness marker | 4 |
| $\mathcal{S}$ | Null space ("soul") | $\ker(\partial Z/\partial t)$ | Impedances immune to all experience | 4 |
| $\beta$ | Murray bifurcation ratio | $2^{1/3} \approx 1.26$ | Optimal branch radius ratio | 1 |
| $\Omega$ | Dual-network overhead | $\geq 1$ | Packing cost of two trees in one volume | 1 |

5. **Paper 5: Clinical Validation** [5]. Proves C2 governs all 29 tissue types (not just neurons); validates against NHANES ($n = 49{,}774$, zero fitted parameters, 7/7 organs significant); reports all-cause mortality $C = 0.70$, Q1-vs-Q4 risk ratio = 6.89×; provides sex-as-boundary-condition analysis.

Papers 1–4 are primarily theoretical with internal computational verification. Paper 5 breaks the circularity with external epidemiological data.

## V. VERIFICATION AND VALIDATION MATRIX

Table II classifies every major claim by its current evidence status. Four levels are used:

**IC: Internal consistency.** Verified within the ALICE simulation engine. Self-confirming; does not constitute independent evidence.

**LIT: Literature-supported.** Prediction matches published experimental or clinical data not used in model construction.

**EXT: Externally validated.** Tested against independent dataset (NHANES) with zero fitted parameters and hash-locked predictions.

**UT: Untested.** Falsifiable prediction not yet confronted with data.

## VI. PREDICTION ROADMAP

Table III organises the falsifiable predictions of the Γ-Net framework by experimental modality, enabling independent groups to test specific claims without needing the full simulation engine.

## VII. THE INTERFACE SCALE TABLE

The same equation (Γ, C1, C2) applies across seven distinct physical domains and 14 orders of magnitude in spatial scale:
At every scale, the physics is identical: only the definition of $Z$ changes (electrical, acoustic, mechanical, hydraulic, thermal, electromagnetic, chemical). The universality is not an analogy; it is the same theorem applied to different media.

Moreover, every row in the table participates in the same closed loop: **perception** ($x_{\text{in}}$ measurement) → **error** (Γ mismatch) → **compensation** (C2: $\Delta Z = -\eta\,\Gamma\, x_{\text{in}}\, x_{\text{out}}$) → **re-perception** (updated $\Gamma'$ and $x'_{\text{in}}$). This perception–compensation cycle is the operational meaning of C1–C2 at every interface; Paper 5 elaborates it across all seven domains.

## VIII. THE TWENTY-NINE C2 NETWORKS

Paper 5 proves that C2 (Eq. 4) recovers 29 independently discovered tissue-remodeling laws when the

TABLE II. Verification and validation matrix. IC = internal consistency; LIT = literature-supported; EXT = externally validated (NHANES); UT = untested prediction.

| **Claim** | **Paper** | **IC** | **LIT** | **EXT** | **UT** |
|---|---|---|---|---|---|
| *Topology and scaling (Paper 1)* | | | | | |
| Relay nodes emerge from MRP | 1 | ✓ | | | |
| $\tau_{\text{conv}} \propto N^{-0.46}$ | 1 | ✓ | | | |
| $A_{\text{cut}} > 0$ (irreducibility) | 1 | ✓ | | | |
| Murray ratio $\beta = 2^{1/3}$ | 1 | ✓ | ✓ | | |
| Kleiber's law $B \propto M^{3/4}$ | 1 | ✓ | ✓ | | |
| Joint = quarter-wave transformer | 1 | ✓ | | | ✓ |
| $A_{\text{cut}} \propto M^{2/3}$ | 1 | ✓ | ✓ | | |
| *Dual-network organ health (Paper 2)* | | | | | |
| Murray's law from $\Gamma$-minimisation | 2 | ✓ | ✓ | | |
| Dual-network health $H = (1-\Gamma_n^2)(1-\Gamma_v^2)$ | 2 | ✓ | | ✓ | |
| Vascular disease as $Z$-mismatch | 2 | ✓ | ✓ | | |
| 12/12 clinical calibration tests | 2 | | ✓ | | |
| *Temporal dynamics (Paper 3)* | | | | | |
| Sleep = $D_Z$ discharge | 3 | ✓ | ✓ | | |
| Jellyfish sleep without brain | 3 | | ✓ | | |
| Brain required for thermal management | 3 | ✓ | ✓ | | |
| Lifecycle bathtub curve | 3 | ✓ | ✓ | | |
| Yerkes–Dodson as impedance optimum | 3 | ✓ | ✓ | | |
| Aging = Arrhenius fatigue on $Z$ | 3 | ✓ | | | |
| Aging pain immunity (slow drift) | 3 | ✓ | | | ✓ |
| Infant sleep duration vs $D_Z$ | 3 | ✓ | ✓ | | |
| *Memory, consciousness, nociception (Paper 4)* | | | | | |
| Memory = $dZ/dt$ | 4 | ✓ | | | |
| $\Phi_{\text{meta}}$ as consciousness marker | 4 | ✓ | | | ✓ |
| Null space $\mathcal{S} = \ker(\partial Z/\partial t)$ | 4 | ✓ | | | |
| Television Theorem (open/closed loop) | 4 | ✓ | | | ✓ |
| Thermal hierarchy: PFC fails before V1 | 4 | ✓ | ✓ | | ✓ |
| *Clinical validation (Paper 5)* | | | | | |
| C2 = 29 tissue remodeling laws | 5 | | ✓ | | |
| NHANES 7/7 organ AUC significant | 5 | | | ✓ | |
| Endocrine AUC = 0.823 | 5 | | | ✓ | |
| All-cause mortality $C = 0.70$ | 5 | | | ✓ | |
| Q1 vs Q4 mortality ratio = 6.89× | 5 | | | ✓ | |
| Sex as boundary condition on $Z_{\text{setpoint}}$ | 5 | | ✓ | ✓ | |
| Cardiac AUC gain +0.166 from vascular input | 5 | | | ✓ | |
| $H$ decreases with disease burden ($\rho = -0.378$) | 5 | | | ✓ | |

generic signals $(x_{\text{in}}, x_{\text{out}})$ are identified with tissue-specific physical quantities. Table V provides the mapping.

## IX. TISSUE BLUEPRINTS: FROM TABLE TO TOPOLOGY

Table V maps C2 to 29 remodeling laws, but it does not specify *how* those tissues are physically wired. This section defines *tissue blueprints*: concrete `GammaTopology` instances in which every node is a `GammaNode` with a biologically assigned mode count $K$, and every edge is a `MulticoreChannel` enforcing C1/C2/C3 at each tick.

A blueprint contains:

1. A set of `GammaNode`s with tissue-appropriate `TissueType` (which sets $K$, $Z_{\text{mean}}$, and myelination).

2. A wiring diagram: directed edges (source, target).

TABLE III. Falsifiable prediction roadmap. Each prediction is testable with existing technology.

| Domain | Prediction | Method |
|---|---|---|
| *Physiological* | | |
| | PFC fails before V1 under hyperthermia | Controlled heating + fMRI/EEG |
| | Vascular $\Gamma$ correlates with PWV / elastography | BIS + arterial tonometry |
| | Joint cartilage $\approx$ quarter-wave transformer | Ultrasound impedance spectroscopy |
| | Bone-on-bone OA raises local $\Gamma^2$ measurably | Intra-articular force sensors |
| *Sleep & circadian* | | |
| | Cross-species $D_{\text{thermal}}/D_{\text{micro}}$ ratio predicts NREM/REM partition | Polysomnography + metabolic calorimetry |
| | Sleep rebound magnitude $\propto$ wake-duration $D_Z$ | EEG slow-wave power after controlled deprivation |
| | Jellyfish sleep duration scales with $\kappa_{\text{eff}}$ | Temperature-controlled aquarium experiments |
| *Cognitive / developmental* | | |
| | Infantile amnesia offset tracks myelination Z-change | Retrospective longitudinal EEG + structural MRI |
| | Yerkes–Dodson peak shifts with $\eta$ (skill level) | Psychophysics + cortisol monitoring |
| | Aging pain immunity: slow $Z$-drift below pain threshold | Quantitative sensory testing, longitudinal |
| *Clinical / epidemiological* | | |
| | $\Gamma$-vector predicts disease in non-US populations | NHANES-equivalent cohorts (UK Biobank, Taiwan MJ) |
| | $H$ predicts surgical outcome / length of stay | Prospective peri-operative cohort |
| | Network propagation: renal $\Gamma$ predicts cardiac events | Longitudinal cardiorenal registry |
| *Thermoregulation / comparative* | | |
| | $T_c(K)$: no high-$K$ organism sustained below $T_c$ with cognition | Comparative physiology database + torpor EEG |
| | Hibernation EEG power scales with $K \cdot \Gamma^2(T)$ | Cross-species torpor polysomnography |
| | Nocturnal shelter-seeking scales with $K$ | Comparative ethology $\times$ thermoregulation data |

TABLE IV. Representative interfaces grouped by physical domain. Full table with impedance definitions: Paper 5 [5].

| Domain | Interface | Scale |
|---|---|---|
| Electrical | Synapse / ion channel | 25 nm–1 $\mu$m |
| Electrical | Power-grid transformer | 10 m |
| Hydraulic | Arteriole (Poiseuille) | 100 $\mu$m |
| Hydraulic | Whole-body vascular | 1 m |
| Acoustic | Tissue boundary (ultrasound) | 1 mm |
| Acoustic | Submarine hull | 10 m |
| Electromagnetic | Optical fibre splice | 100 $\mu$m |
| Electromagnetic | Ionospheric waveguide | $10^6$ m |
| Mechanical | Bone–cartilage joint | 10 cm |
| Mechanical | Tendon–muscle junction | 1 cm |
| Thermal | Brain thermal boundary | 10 cm |
| Thermal | Skin–air interface | 1 m |
| Chemical | Organ parenchyma | 10 cm |

3. A factory function returning a ready-to-run `GammaTopology`.

Diseases are *not coded* in the blueprint. Disease = perturbation of initial conditions (impedance shift, edge removal, or stimulus pattern) applied to a healthy blueprint, then evolved via `GammaTopology.tick()`.

### A. Cardiovascular blueprint

Eleven nodes span four $K$-levels:

| Node | TissueType | $K$ |
|---|---|---|
| SA node, AV node, LV, RV, aortic valve, pulm. valve, aorta, coronary | Cardiac Purkinje | 4 |
| Arterioles | Vasc. smooth muscle | 2 |
| Pulmonary artery | Autonomic preganglionic | 2 |
| Capillaries | Endothelial | 1 |

Fourteen directed edges connect the conduction system (SA→AV→LV/RV), left-heart systemic path (LV→valve→aorta→coronary/arterioles→capillaries), right-heart pulmonary path, and return paths. The dimensional cascade $K{=}4 \to 2 \to 1$ guarantees $A_{\text{cut}} > 0$ (Paper 1 irreducibility theorem), physically representing the loss of pulsatile modes as blood flows from elastic arteries to resistance vessels.

### B. Neural blueprint

Nine nodes span four $K$-levels:

TABLE V. The 29 Γ-networks. Each row maps C2 to a known remodeling law by identifying $x_{\text{in}}$, $x_{\text{out}}$, and $Z$. Full derivations: Paper 5 [5].

| **Tissue** | **Known law** | $x_{\text{in}}$ | $x_{\text{out}}$ | $Z$ **definition** |
|---|---|---|---|---|
| Neural | Hebb's law (1949) | Pre-synaptic spike | Post-synaptic EPSP | $R_m + 1/j\omega C_m$ |
| Vascular | Glagov remodeling (1987) | Wall shear stress | Endothelial response | $P/Q$ (Poiseuille) |
| Skeletal | Wolff's law (1892) | Applied load | Bone strain | Force/velocity |
| Muscular | Davis's law (1867) | Neural drive | Muscle tension | Stiffness |
| Immune | Clonal selection | Antigen (APC) | T/B cell activation | Receptor affinity |
| Epithelial | Keratinisation | External friction | Tissue deformation | Thermal+mechanical $Z$ |
| Pulmonary | Altitude acclimatisation | Ventilation demand | Alveolar expansion | $P_{\text{aw}}/\dot{V}$ |
| GI tract | Intestinal adaptation | Luminal content | Epithelial uptake | Absorption $Z$ |
| Endocrine | Receptor regulation | Hormone conc. | Intracellular resp. | $1/K_d$ |
| Lymphatic | Reactive hypertrophy | Tissue load | Lymph node response | Drainage $Z$ |
| Renal | TG feedback | Macula densa signal | Arteriolar response | $P_{\text{glom}}/Q_{\text{filt}}$ |
| Hepatic | Liver regeneration | Portal HGF | Hepatocyte prolif. | Clearance $Z$ |
| Connective | Tendon remodeling | Mechanical strain | Fibroblast response | $EA/L$ |
| Adipose | Leptin signaling | Insulin signal | Adipocyte response | Storage $Z$ |
| Hematopoietic | Erythropoiesis | EPO signal | Erythroid response | $P_{O_2}/Q_{O_2}$ |
| Reproductive | Follicular selection | Gonadotropin | Granulosa/germ cell | $1/(N_R K_a)$ |
| Cardiac | Laplace–Starling | Wall stress $\sigma$ | Myocardial strain $\varepsilon$ | Chamber compliance |
| Brain (reg.) | Baroreflex resetting | Sensory afferent | Motor efferent | Controller gain |
| Ocular | IOP homeostasis | Aqueous pressure | Trabecular drainage | $P_{\text{IOP}}/Q_{\text{aq}}$ |
| Auditory | Tonotopic plasticity | Sound pressure | Hair cell response | Cochlear $Z$ |
| Vestibular | VOR adaptation | Angular velocity | Oculomotor response | Vestibular gain |
| Olfactory | Receptor turnover | Odorant conc. | ORN response | Receptor $1/K_d$ |
| Dental | Tertiary dentinogenesis | Thermal/mech. insult | Odontoblast resp. | Dentin $Z$ |
| Enteric | Enteric plasticity | Luminal distension | ENS firing | Gut wall compliance |
| Autonomic | Sympathovagal remod. | Baroreceptor input | Effector output | Sympathovagal $Z$ |
| Pancreatic | $\beta$-cell compensation | Glucose load | Insulin secretion | $1/(N_\beta \cdot k_{\text{sec}})$ |
| CSF | Choroid regulation | CSF pressure | Production rate | $P_{\text{CSF}}/Q_{\text{CSF}}$ |
| Brain (func.) | Cortical map plasticity | Afferent frequency | Cortical area | Synaptic $Z$ matrix |
| Cartilage | *C2 null* ($\eta \approx 0$) | Mechanical load | (negligible) | $H_A A/h$ |

| **Node** | **TissueType** | $K$ |
|---|---|---|
| Cortex (motor, sensory), thalamus | Cortical pyramidal | 5 |
| Spinal (motor, sensory), motor neuron, sensory neuron | Motor $\alpha$ / Sensory A$\beta$ | 3 |
| Nociceptor A$\delta$ | Pain A$\delta$ fibre | 2 |
| Nociceptor C | Pain C fibre | 1 |

Twelve directed edges form descending motor (cortex→thalamus→spinal→motor neuron), ascending sensory (sensory neuron→spinal→thalamus→cortex), pain pathway (C fibre→A$\delta$→spinal), and sensorimotor cross-connections. The cascade $K$=5 → 3 → 2 → 1 defines the *neural cable degradation spectrum*:

| $\Gamma$ | **Clinical correlate** |
|---|---|
| $\approx 0.0$ | Normal conduction (healthy match) |
| $\approx 0.1$ | Fever, neuroinflammation (thermal noise ↑) |
| $\approx 0.3$ | PTSD (structurally locked $\varepsilon$; dormant until triggered, Paper |
| $\approx 0.7$ | ALS (progressive channel collapse) |
| $\to 1.0$ | Phantom limb pain (open-circuit echo) |

These are not disease models; they are predictions of what Γ-values *should* emerge from topology perturbations alone. Paper 1 verifies the topology-level physics; Paper 5 verifies the disease-level emergence.

### C. Complete tissue atlas

The cardiovascular and neural blueprints above are two of 29 organ-system blueprints implemented in `BLUEPRINT_REGISTRY`. Table VI lists all 29, totalling ∼ 217 nodes and ∼ 242 directed edges. Every blueprint

passes C1 energy conservation at every edge and C2 physics verification ($A_{\rm imp}$ non-increasing, $A_{\rm cut}$ invariant) under 100 ticks of evolution.

TABLE VI. Complete tissue blueprint atlas. Nodes ($N$), edges ($E$), and mode count range are listed for each organ system.

| **System** | $N$ | $E$ | $K$-range | **Key feature** |
|---|---|---|---|---|
| Cardiovascular | 11 | 13 | 1–4 | Pulsatile cascade |
| Neural | 9 | 11 | 1–5 | Cable degradation |
| Pulmonary | 7 | 7 | 1–3 | Gas exchange |
| Renal | 9 | 9 | 1–3 | TG feedback |
| Hepatic | 7 | 7 | 1–3 | Dual blood supply |
| Gastrointestinal | 8 | 10 | 1–3 | Mucosal barrier |
| Endocrine | 8 | 11 | 1–3 | Negative feedback |
| Immune | 7 | 9 | 1–3 | Innate/adaptive |
| Lymphatic | 7 | 7 | 1–2 | One-way drainage |
| Musculoskeletal | 7 | 8 | 1–3 | Motor unit |
| Skeletal | 6 | 7 | 1–2 | Wolff's law |
| Epithelial | 8 | 8 | 1–2 | Barrier layers |
| Connective | 6 | 6 | 1–2 | ECM turnover |
| Adipose | 6 | 7 | 1–2 | Metabolic storage |
| Hematopoietic | 9 | 9 | 1–3 | EPO feedback |
| Cartilage | 4 | 4 | 1 | $\eta \approx 0$ (avascular) |
| Reproductive | 6 | 7 | 1–2 | Hormonal axis |
| Brain regulatory | 7 | 8 | 2–3 | Autonomic control |
| Vascular | 11 | 11 | 1–3 | Arterial tree |
| Ocular | 9 | 9 | 1–3 | IOP regulation |
| Auditory | 7 | 7 | 1–3 | Tonotopic cascade |
| Vestibular | 6 | 6 | 1–2 | Balance loops |
| Olfactory | 4 | 4 | 1 | Receptor turnover |
| Dental | 7 | 7 | 1–2 | Pulp–dentin axis |
| Enteric | 7 | 7 | 1–3 | Gut nervous system |
| Autonomic | 8 | 10 | 1–3 | Sympathetic/para |
| Pancreatic | 7 | 8 | 1–2 | Paracrine islet |
| CSF | 8 | 8 | 1–2 | Choroid–ventricle |
| Brain functional | 12 | 21 | 3–5 | Cortex–subcortex |

Cartilage ($\eta \approx 0.001$) and olfactory receptors ($K = 1$) represent the extremes: cartilage barely remodels (avascular), while olfactory neurons are among the few that regenerate throughout life. Brain functional topology has the richest dimensional structure ($K = 5$ cortex vs. $K = 3$ subcortex, 21 edges), producing the highest $A_{\rm cut}$ of any blueprint.

### D. Whole-body inter-organ topology

The 29 blueprints above are *intra*-organ networks. A living body also requires *inter*-organ communication: every organ receives arterial blood supply (vascular bus) and autonomic innervation (neural bus), and the brain serves as a central command hub.

We unify the three chains into a single `GammaTopology` via `build_whole_body()`:

1. **Organ layer:** all 29 blueprints are instantiated with prefixed node names (e.g. `cv.sa_node`, `pulm.alveolus`).

2. **Vascular bus:** an aortic trunk node ($K = 4$) fans out to 29 arteriolar branch nodes ($K = 2$), each connecting to the organ's designated vascular interface node. K-cascade: $4 \to 2 \to K_{\rm organ}$.

3. **Neural bus:** an autonomic trunk node ($K = 3$) fans out to 29 branch nodes ($K = 2$), each connecting to the organ's neural interface node. K-cascade: $3 \to 2 \to K_{\rm organ}$.

4. **Brain hub:** the brain_functional blueprint's `hypothalamus_b` ($K$=3) drives the neural bus trunk (descending command), while the vascular bus feeds back to the brain (interoception).

The resulting whole-body graph contains **278 nodes** and **426 directed edges**, with $K$ spanning 1 to 5. All edges satisfy C1 ($\Gamma^2 + T = 1$) by construction. The vascular bus contributes $A_{\rm cut} = 2$ at every trunk-to-branch edge ($K$=4 $\to$ $K$=2), the neural bus contributes $A_{\rm cut} = 1$ ($K$=3 $\to$ $K$=2), and further cutoff costs accumulate at every bus-to-organ junction. The total structural action $A_{\rm cut} = 222$ is irreducible under all C2 gradients—it is the geometric price of multi-organ life.

Disease cascades emerge automatically from this topology: raising the aortic impedance (vascular stenosis) increases $\Gamma$ at all downstream branches, which propagates via the bus to every organ's vascular interface. No disease-specific code is needed (E0). The 145 whole-body tests (`tests/test_whole_body.py`) verify structure, C1/C2 physics, bus connectivity, dimensional cascade, signal propagation, and reproducibility.

### E. Thermoregulation as a Γ-Net necessity

Blood is the transmission medium of the vascular bus. Its viscosity $\eta$ depends on body temperature $T$ via the Arrhenius relation:

$$\tilde{\eta}(T) \;=\; \exp\Big[\alpha\Big(\frac{1}{T} - \frac{1}{T_0}\Big)\Big], \qquad \alpha \equiv \frac{E_a}{R} \approx 1949\ \mathrm{K}, \quad (5)$$

where $E_a \approx 16.2\ \mathrm{kJ\,mol^{-1}}$ is the activation energy for viscous flow in plasma and $T_0 = 310$ K (37 °C). In the Poiseuille limit, the vascular bus impedance $Z_{\rm bus}(T) = Z_{\rm ref}\,\tilde{\eta}(T)$. At every organ–bus junction the reflection coefficient becomes

$$\Gamma_i(T) \;=\; \frac{Z_{{\rm organ},i} - Z_{\rm bus}(T)}{Z_{{\rm organ},i} + Z_{\rm bus}(T)}\,, \quad (6)$$

and the vascular contribution to the whole-body action is $A_{\rm vasc}(T) = \sum_i K_{{\rm common},i}\,\Gamma_i^2(T)$.

C2 repair runs at a rate that is itself temperature-dependent (enzymatic Arrhenius, $\tilde{\eta}_{\rm learn}(T) = \exp[\beta(T_0^{-1} - T^{-1})]$ with $\beta = E_{\rm learn}/R \approx 6015$ K). The repair time therefore scales as

$$\tilde{\tau}(T) \;=\; \frac{\tilde{\eta}(T)}{\tilde{\eta}_{\rm learn}(T)} \;=\; \exp\Big[(\alpha+\beta)\Big(\frac{1}{T} - \frac{1}{T_0}\Big)\Big], \qquad (7)$$

which is *doubly exponential* in $1/T$: viscosity rises while repair rate falls.

Setting $\tilde{\tau}(T_c) = \tau_{\max}$ yields the **critical temperature**

$$T_c \;=\; \frac{T_0}{1 + \dfrac{T_0\,\ln\tau_{\max}}{\alpha+\beta}}\,. \qquad (8)$$

For an organism with brain modal complexity $K$, the action scales as $K\,\Gamma^2$, so the effective $\tau_{\max}$ shrinks by $1/K$, giving $T_c(K) > T_c(1)$ for $K > 1$ (dimensional amplification). The 93 thermoregulation tests confirm:

$$T_c(K{=}1) = -10.3\,^\circ\!{\rm C}\ \text{(ectotherm viable)}$$
$$T_c(K{=}5) = +4.5\,^\circ\!{\rm C}\ \text{(endothermy required)}$$

Higher $K$ brains *must* maintain $T > T_c(K)$, rendering endothermy a necessary condition for complex neural topology—not a metabolic luxury but a $\Gamma$-topological constraint.

*a. Thermodynamic boundary crisis during $D_Z$ clearance.* To discharge accumulated impedance debt $D_Z$ (Paper 3 [3]), the system must enter a low-power state (sleep), reducing metabolic heat generation to a basal level $P_{\rm basal}$. System temperature during sleep obeys

$$T_{\rm sys} = T_{\rm env} + P_{\rm basal}\,R_{\rm th}\,, \qquad (9)$$

where $R_{\rm th}$ is the organism's effective thermal resistance (integrating fur, blubber, posture, and huddling—all abstracted as a single lumped parameter). The C2 boundary condition demands $T_{\rm sys} > T_c(K)$ at all times; violation halts remodelling and freezes impedance mismatch in place.

During nocturnal or seasonal temperature drops, $T_{\rm env}$ may swing by tens of kelvin. When $P_{\rm basal}\,R_{\rm th}$ alone cannot guarantee the inequality, the organism must couple to an *environmental thermal low-pass filter*—any high-heat-capacity mass (subterranean rock, tree hollows, snow caves, or conspecific huddles) whose thermal inertia damps the external oscillation:

$$T_{\rm refugium}(z) \approx \overline{T}_{\rm annual}\,, \qquad z \gg \delta = \sqrt{2\kappa/(\rho c_p\,\omega)}\,, \qquad (10)$$

where $\delta$ is the thermal penetration depth. Ecologically, such environments are called *thermal refugia*; physically, they are low-pass filters that replace the volatile $T_{\rm env}$ with a stable baseline $\overline{T}_{\rm annual}$.

The resulting constraint chain is purely physical: high $K \to$ high $T_c(K) \to$ sleep reduces $P_{\rm met} \to P_{\rm basal}\,R_{\rm th}$ may be insufficient $\to$ external thermal coupling required. Low-$K$ organisms ($T_c \ll 0\,^\circ\!{\rm C}$) satisfy the inequality with biological insulation alone and need no refugium.

*b. Falsification criterion.* The $T_c(K)$ formula is falsified by exhibiting an organism with (i) closed high-pressure vasculature, (ii) $K \geq 5$, (iii) sustained $T_{\rm body} \leq T_c(K)$, and (iv) maintained high-level cognition. No such organism is currently known. Mammalian hibernators drop $T$ below $T_c$ but simultaneously reduce brain activity to torpor [7, 8]; ectotherms with complex behaviour operate on the low-$K$ branch where $T_c$ is low [9]; cross-vertebrate data confirm that species with higher, more stable body temperatures develop larger brains at given body mass [10, 11].

*c. Gating, "willpower," and time perception.* The preceding analysis explains winter lethargy as a minimum-action consequence: at low $T$, $\Gamma^2$ rises on every channel, and the cheapest response is quiescence—close thalamic gates ($G_i \to 0$; see the television inequality, Paper 4 [4]) and reduce the summation $A = \sum_i K_i\,\Gamma_i^2\,P_{{\rm in},i}$. This is the physics behind "not wanting to move in winter."

Yet organisms override it. The mechanism is *bandwidth reallocation under a fixed thermal ceiling* $P_{\max}$. By closing $N-1$ thalamic gates the prefrontal cortex receives only one stream; its full processing quality $Q_{\rm PFC}$ and bandwidth $B_{\rm PFC}$ are devoted to a single task. The temporal resolution of the attended channel improves from $\Delta t \sim N/B_{\rm PFC}$ (multiplexing $N$ streams) to $\Delta t \sim 1/B_{\rm PFC}$ (single stream). What folk psychology calls "willpower" is this gating reallocation: a change in the *functional impedance* $Z_{\rm fun}$ (gate gains $G_i$, arousal $Z_{\rm arousal}$), not in the structural impedance $Z_s$ that encodes long-term patterns. The subjective effort is real—the active channel's $\Gamma^2 P_{\rm in}$ at low $T$ exceeds its cost at $T_0$, and the $N-1$ silenced channels surrender all fidelity—but the *structural* network is unchanged. This is why "willpower" cannot rewrite deeply forged patterns: gating redistributes bandwidth; only C2 remodelling ($\Delta Z_s = -\eta\,\Gamma\,x_{\rm in}\,x_{\rm out}$) can reshape the impedance itself (Paper 3, Remark 4 [3]; Paper 4, Remark 3 [4]). Paper 4 further decomposes effective willpower into three multiplicative factors—metabolic budget $\eta_{\rm pfc}$, gate quality $Q_{\rm gate}$, and memory calibration $(1 - \langle\Gamma^2_{\rm memory}\rangle)$—so that a single zero factor extinguishes volitional capacity regardless of the other two [4].

The extreme case is emergency arousal. In a car crash, the brainstem drives $Z_{\rm arousal}$ to its ceiling and the thalamus gates every non-threat modality to zero. The PFC processes a single high-$K$ stream at maximum bandwidth; temporal resolution increases by a factor $\sim N$ and the event is perceived in *slow motion*—not an illusion, but a real $N$-fold increase in sampling rate (Paper 4, Theorem 1 [4]). Conversely, fatigue and cold *reduce* $B_{\rm PFC}$ (higher $\Gamma^2$), so fewer "frames" are processed per second and time feels faster—the well-known subjective acceler-

ation of uneventful cold afternoons.

### F. Emergence classification

The operational status of every disease model in the codebase is classified by *emergence level*:

| Level | Name | Definition |
|---|---|---|
| E0 | True Emergence | Only C1/C2/C3 + initial conditions. Zero disease-specific code. |
| E1 | Parameterised | C2 via `ImpedanceChannel.remodel()` but requires per-disease params. |
| E2 | Scripted | Ad-hoc $Z$ updates. Legacy only. |

**Rule**: all new disease models must be E1 or higher. E2 is forbidden for new code. E0 is the target: if behaviour emerges from C2 + initial conditions alone, no additional parameters should be added.

The tissue blueprints are E0 by construction: the `GammaTopology` enforces C1 at every edge, applies C2 remodeling at every tick, and the only inputs are the initial $Z$-vectors and the perturbation (impedance shift or edge removal).

Source code: `alice/body/tissue_blueprint.py` (29 factory functions in `BLUEPRINT_REGISTRY`), `tests/test_tissue_blueprints.py` (245 parametrised tests covering structure, C1, C2, and organ-specific physiology for all 29 blueprints), plus the E0 emergence suites `tests/test_e0_cardiovascular.py`, `tests/test_e0_neural.py`, and `tests/test_e0_metabolic.py` (56 tests total). All pass with 100% rate.

#### 1. *Ablation structure and epistemic credibility*

A test suite written alongside the code it tests is vulnerable to circular validation: the same author who writes `simulate_diabetes()` also writes `assert result == 0.42`, creating a self-confirming closed loop whose pass count carries no epistemic weight.

We address this by imposing a *structural ablation constraint* on every E0 test file. Each file contains two mutually contradictory test classes that must pass simultaneously:

**C2-on class:** ($\eta > 0$, impedance remodeling active): asserts that (i) the action functional $A[\Gamma]$ decreases over time, (ii) pathological perturbations are partially compensated, and (iii) different initial conditions produce distinguishable $\Gamma$-trajectories.

**C2-off class:** ($\eta = 0$, remodeling disabled): asserts that (i) $A[\Gamma]$ does *not* systematically decrease, (ii) impedance perturbations persist permanently ($Z_{\text{after}} = Z_{\text{before}}$ to machine precision), and (iii) the total action exceeds the C2-on case.

The logical structure is a pincer: a hardcoded implementation that fakes "action always decreases" will pass the C2-on class but fail the C2-off class (which requires action to persist). A hardcoded implementation that fakes "action stays constant" will pass the C2-off class but fail the C2-on class. The *only* implementation that simultaneously satisfies both classes is one that actually computes $\Delta Z = -\eta\,\Gamma\,x_{\text{in}}\,x_{\text{out}}$ and propagates its consequences through the topology.

This does not replace external validation (Table II distinguishes IC from LIT and EXT), but it raises the internal-consistency evidence above the level of trivially satisfiable assertions. Among the full test suite of 3,797 automated tests, approximately 65–75% verify hardcoded values, boundary clamping, or algebraic identities that any implementation could trivially pass. The ablation-structured E0 tests constitute a smaller fraction ($\sim$56 tests across three topologies), but they carry disproportionate epistemic weight because they are non-trivially satisfiable.

## X. WHAT THIS THEORY DOES NOT CLAIM

The $\Gamma$-framework derives *physical constraints* that any living system must satisfy. It does *not* reconstruct evolutionary history. Specifically:

1. **Necessary conditions, not sufficient causes.** The framework shows that a $K$=5 brain *must* maintain $T > T_c(5)$, *must* have vasculature, and *must* seek thermal shelter during sleep. It does not explain *why* such brains appeared or *which* lineages developed them.

2. **No evolutionary trajectory.** Statements such as "the tidal zone offers the lowest-action path" identify geometric optima in impedance space. Whether any lineage actually traversed that path is a question for palaeontology and phylogenetics, not for this framework.

3. **C2 uniqueness is mathematical, not biological.** The proof that C2 is the unique first-order $\mathcal{A}$-minimising rule (Paper 5) is a theorem about gradient descent. That multiple molecular machineries converge on the same equation is *consistent with* C2 being a physical attractor; it is not evidence that evolution "discovered" C2.

4. **Compatibility, not replacement.** Natural selection determines which lineages survive; the $\Gamma$-framework supplies the physical objective function that selection operates on. The two are complementary, not competing.

Throughout the six papers, any sentence referencing evolution, convergence, or the Cambrian radiation should be read as *physical constraint language*: "$X$ is required" means "any system with property $Y$ must satisfy $X$"; it does not mean "$X$ evolved because of $Y$".

## XI. NOTATION CONVENTIONS

Table VII collects notation conventions used across Papers 0–5. For complete definitions including derivations, see the quantity inventory (Table I).

TABLE VII. Notation conventions used across all six papers.

| Symbol | Meaning | Units / Range |
|---|---|---|
| $Z$ | Impedance (effort/flow) | $\Omega$; Pa·s/m$^3$; N·s/m |
| $\Gamma$ | Reflection coefficient | dimensionless, $[-1, +1]$ |
| $T$ | Transmission coefficient $(1-\Gamma^2)$ | dimensionless, $[0, 1]$ |
| $\eta$ | Remodeling rate | $> 0$; $\approx 0$ for cartilage |
| $(x_{\text{in}},\, x_{\text{out}})$ | Interface signals | signal units |
| $\mathcal{A}$ | Reflection action | J·s |
| $A_{\text{imp}}$ | Improvable action | J·s |
| $A_{\text{cut}}$ | Cutoff (irreducible) action | J·s |
| $K$ | Modal complexity | integer $\geq 1$ |
| $K_{\text{com}}$ | Common modes $\min(K_i, K_j)$ | integer |
| $Z_v,\ Z_n$ | Vascular / neural impedance | Pa·s/m$^3$; $\Omega$ |
| $D_Z$ | Impedance debt | J·s |
| $T_c(K)$ | Critical temperature | °C |
| $\beta_{\text{recal}}$ | N3 recalibration rate | s$^{-1}$ |
| $\Phi_{\text{meta}}$ | Meta-reflection coefficient | dimensionless, $[-1, +1]$ |
| $\varepsilon$ | Memory seal factor | dimensionless, $> 0$ |
| $\theta_{\text{recall}}$ | Recall SNR threshold | dimensionless |
| $\mathcal{S}$ | Null space ("soul") | — |
| $H$ | Health index $\prod_i (1-\Gamma_i^2)$ | dimensionless, $[0, 1]$ |

## XII. CONCLUSION

The Γ-Net framework rests on one variational principle (MRP) and three local constraints (C1: energy conservation; C2: impedance remodeling; C3: impedance-tagged transport). From these, five companion papers derive topology, organ health, temporal dynamics, consciousness, and clinical diagnostics—spanning 15 orders of spatial magnitude and 29 tissue types, now unified into a single whole-body topology (278 nodes, 426 edges) through vascular and neural inter-organ buses, with a rigorous thermoregulation derivation showing that endothermy is a Γ-topological necessity for brains above a critical modal complexity $K$.

This paper provides the master reference: a canonical quantity list (Table I), a verification matrix distinguishing internal from external evidence (Table II), and a falsifiable prediction roadmap (Table III).

All source code, simulation scripts, and NHANES pipelines are available at `https://github.com/cyhuang76/alice-gamma-net` under AGPL-3.0. The full test suite comprises 3,797 automated tests with 100% pass rate, of which 56 are ablation-structured E0 emergence tests (Sec. IX F 1).


[1] H.-Y. Huang, "Topology and fractal cost of Γ-networks: relay nodes, metabolic scaling, and the irreducibility theorem," companion Paper 1 (2026).

[2] H.-Y. Huang, "Dual-network organ health: vascular Γ-physics and the five failure modes," companion Paper 2 (2026).

[3] H.-Y. Huang, "Temporal dynamics of impedance debt: sleep, the lifecycle equation, and the physics of aging," companion Paper 3 (2026).

[4] H.-Y. Huang, "Memory, consciousness, and nociception as impedance dynamics," companion Paper 4 (2026).

[5] H.-Y. Huang, "The grand unification: Γ as the universal interface law from microtubules to power grids, and C2 as universal remodeling in twenty-nine tissues," companion Paper 5 (2026).

[6] H.-Y. Huang, *ALICE: A physics-driven electronic lifeform*, `https://github.com/cyhuang76/alice-gamma-net` (2026).

[7] H. Carey, M. Andrews, and S. Martin, "Mammalian hibernation: cellular and molecular responses to depressed metabolism and low temperature," *Physiol. Rev.* **83**, 1153 (2003); see also PMC6674705.

[8] S. Giroud *et al.*, "The torpid state: recent advances in metabolic adaptations and protective mechanisms," *Front. Physiol.* **11**, 623 (2020); PMC6629779.

[9] F. Seebacher, "The evolution of metabolic regulation in animals," *Comp. Biochem. Physiol. B* **224**, 207 (2024); PMC11880559.

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[11] A. Yu *et al.*, "Brain size is predicted by the intensity of intrasexual competition in birds," *Ecol. Evol.* **11**, 2652 (2021); PMC7907648.